

Dongqi Lin, Ph. D.

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RESEARCH PROFILE

I am an atmospheric scientist whose research advances our understanding of the dynamics of high-impact weather by developing physics-based frameworks that connect atmospheric dynamics across spatial and temporal scales to hazard-relevant processes. Currently 3.5 years post-PhD, I have established an independent and internationally recognised research trajectory in atmospheric science. My work integrates high-resolution modelling, complex observational analysis, and interdisciplinary methods.

EDUCATION

- Apr. 2019 - Ph. D. in Physics, University of Canterbury (UC), New Zealand
Oct. 2022
- Sep. 2017 - MRes in Climate and Atmospheric Science, University of Leeds, United Kingdom
Oct. 2018 (Distinction)
- Sep. 2016 - BSc in Applied Mathematics and Meteorology, University of Reading, United Kingdom
Jun. 2017 (1st Class Honours)
- Sep. 2013 - BSc in Mathematics and Applied Mathematics, Nanjing University of Information Science
Jun. 2017 and Technology (NUIST), China (average grade of 87%)

PROFESSIONAL APPOINTMENTS

- Apr. 2024 - Research Fellow, the ARC Centre of Excellence for Climate Extremes (CLEX), Monash
Present University, Melbourne, VIC, Australia
- Aug. 2022 - Postdoctoral Fellow, University of Canterbury, Christchurch, New Zealand
Mar. 2024

PUBLICATIONS

- Lin, D.**, Lin, D., Saunders, K.R. and Singh, M.S., 2026. Mapping the Spatial Scales of Australian Extreme Precipitation Using Daily Rain Gauges. *International Journal of Climatology*, p.e70283..
- Lin, D.**, Katurji, M., Zavar-Reza, P., Malyarenko, A., Valencia, A. and Zhang, J., 2025. High-fidelity experimental simulations of atmospheric flow near low-intensity fires in a wildland-urban interface. *Journal of Geophysical Research: Atmospheres*, 130(19), p.e2025JD043641.
- Valencia A., Penney G., Baker G., Gorham D., Gill F., **Lin, D.**, Katurji M., and Power A. 2024. Development of wildland fire exposure indicator for performance-based design applications. iCFSERP20230170. [Peer-reviewed conference proceeding]
- Lin, D.**, Zhang, J., Khan, B., Katurji, M. and Revell, L.E., 2023. GEO4PALM v1. 1: an open-source geospatial data processing toolkit for the PALM model system. *Geoscientific Model Development Discussions*, 17, 815-845, <https://doi.org/10.5194/gmd-17-815-2024>, 2024.
- Bhatti, Y. A., Revell, L. E., McDonald, A. J., Archibald, A. T., Schuddeboom, A. J., Williams, J., Hardacre, C., Mulcahy, J., and **Lin, D.**, Aerosol and Dimethyl Sulfide Sensitivity to Global Sulfate Chemistry Schemes, *Journal of Geophysical Research: Atmospheres*, 129, e2023JD040635. <https://doi.org/10.1029/2023JD040635>, 2024.
- Lin, D.**, Katurji, M., Revell, L. E., Khan, B., and Sturman, A.: Investigating multiscale meteorological controls and impact of soil moisture heterogeneity on radiation fog in complex terrain using semi-idealised simulations, *Atmospheric Chemistry and Physics*, 23, 14451-14479, <https://doi.org/10.5194/acp-23-14451-2023>, 2023.
- Lin, D.**, Katurji, M., Revell, L. E., Khan, B., Osborne, N., Soltanzadeh, I., and Kremser, S: Fog type classification using a modified Richardson number for Christchurch, New Zealand. *International Journal of Climatology*, 43(1), 314- 330. <https://doi.org/10.1002/joc.7761>, 2023.
- Nathan, B., Kremser, S., Mikaloff-Fletcher, S., Bodeker, G., Bird, L., Dale, E., **Lin, D.**, Olivares, G., and Somervell, E.: The MAPM (Mapping Air Pollution eMissions) method for inferring particulate matter emissions maps at city scale from in situ concentration measurements: description and demonstration of capability, *Atmospheric Chemistry and Physics*, 21, 14089-14108, <https://doi.org/10.5194/acp-21-14089-2021>, 2021.

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- Lin, D.**, Khan, B., Katurji, M., Bird, L., Faria, R., and Revell, L. E.: WRF4PALM v1.0: a mesoscale dynamical driver for the microscale PALM model system 6.0, *Geoscientific Model Development*, *14*, 2503-2524, <https://doi.org/10.5194/gmd-14-2503-2021>, 2021.
- Lin, D.**, Pickering, B., and Neely III, R. R.: Relating the Radar Bright Band and Its Strength to Surface Rainfall Rate Using an Automated Approach, *Journal of Hydrometeorology*, *21*(2), 335-353. <https://doi.org/10.1175/JHM-D-19-0085.1>, 2020.

Manuscript in preparation

- Vishwanathan G., McDonald A. J., Stone D. A., Rosier S., Noble C., Patterson M., **Lin D.**, and A. Thomas, The role of atmospheric blocking in extreme precipitation events in Aotearoa New Zealand, *Journal of Geophysical Research: Atmospheres* (in review)

AWARDS AND HONORS

- Jun. 2025 Advancing Diversity in Science Grant (AUS \$10,000), Faculty of Science, Monash University
- Sep. 2024 IOP publishing Trusted Reviewer
- Oct. 2023 UC Library Open Access Fund (NZD \$4500)
- 2019 - 2022 UC connect doctoral scholarship (NZD \$21,000 stipend + fees) collaborating with the MBIE project: Mapping Air Pollution eMissions
- Sep. 2022 UC Foundation Doctoral Publication Prize
- Nov. 2021 Best presentation (3rd year student), School of Physical and Chemical Sciences Postgraduate Annual Symposium, University of Canterbury
- May 2021 UC Foundation Doctoral Publication Prize
- Nov. 2020 Best student talk, New Zealand MetSoc Annual Conference
- Jul. 2014 Scholarship for Geographic Information System (GIS) training (CNY 20,000), Saint Louis University, St. Louis, Missouri, United States
- 2013 - 2017 The Third-Class Scholarship awarded by NUIST, Nanjing, China

CONFERENCES AND INVITED TALKS

- Pletzer, T., **Lin, D.**, Zawar-Reza, P., and Katurji, M.: Revealing Hidden Coherent Structures in Plume-Atmospheric Dynamics, The 37th international conference on Alpine Meteorology, Croatia, 2025.
- Lin, D.**, Saunders, K. R., and Singh, M. S.: Do the environments for large-scale and small-scale precipitation extremes differ? The 31st conference of Australian Meteorological and Oceanographic Society (AMOS), Cairns, Australia, 2025
- Lin, D.**, Saunders, K. R., and Singh, M. S.: Do the environments for large-scale and small-scale precipitation extremes differ? Climate and Ocean: Variability, Predictability and Change (CLIVAR), Lorne, Australia, 2025.
- Lin, D.**, Katurji, M., Zhang, J., Malyarenko, A., and Valencia A.: Fire-atmosphere dynamics in a rural-urban interface using turbulence resolving simulations. AGU Fall meeting, San Francisco, 2023.
- Le Cap, C., Carlier, J., Katurji, M., **Lin, D.**, Quénol, H., Georgeault, P., Buisson, E., and Heitz, D.: Towards numerical simulation of a wind machine during spring frost calibrated with field measurements, EGU General Assembly 2023, Vienna, Austria, 2023.
- Lin, D.**, Katurji, M., and Purdie, H.: Impact of crevasses on surface energy balance at an alpine glacier, EGU General Assembly 2023, Vienna, Austria, 2023.
- Lin, D.**, Katurji, M., Valencia, A., and Gill, F.: Fire-atmosphere dynamics at a rural-urban interface using turbulence-resolving meteorological simulations, EGU General Assembly 2023, Vienna, Austria, 2023.
- Katurji, M., Zhang, J., Valencia, A., **Lin, D.**, Strand, T., Pearce, G., Finney, M., Clements, C., and Gross, S.: Atmosphere and fire interactions from the New Zealand experimental burn campaigns, EGU General Assembly 2023, Vienna, Austria, 2023.
- Valencia, A., Katurji, M., **Lin, D.**, Gross, S., Zhang, J., Pearce, G., Finney, M., and Strand, T.: Unravelling wildland fire morphology and structure using image velocimetry, EGU General Assembly 2023, Vienna, Austria, 2023.
- Lin, D.**, Weather Research and Forecasting (WRF) model for air pollution applications, Clean Air Society of Australia and New Zealand (CASANZ) workshop, online, 2023. [Invited talk]
- Lin, D.**, Katurji, M., Khan, B., Revell, L.E.: Fog dynamics and predictability using a coupled urban-atmosphere numerical model for Christchurch, New Zealand, 13th International Conference on Southern Hemisphere Meteorology and Oceanography (ICSHMO), online, 2022.
- Lin, D.**, Khan, B., Katurji, M., Bird, L., Faria, R., and Revell, L.: WRF4PALM v1.0: A Mesoscale Dynamic Driver for the Microscale PALM Model System 6.0, EGU General Assembly 2021, online, 2021.

- Bowery, A., Sparrow, S., Wallom, D., Carver, G., Köhler, M., Weischeimer, A., **Lin, D.**, Katurji, M., and Rosier, S.: Expanding the climateprediction.net project: the creation of a multi model distributed computing infrastructure for climate science. Academia Sinica Grid Computing Centre (ASGC), Taiwan (online), 2021.
- Lin, D.**, Pickering, B., and Neely III, R. R.: Identifying the Microphysical Difference in UK Precipitation for Periods with and without Radar Bright Band. National Facilities for Atmospheric and Radar Research (NFARR) User Meeting, Chilbolton, United Kingdom, 2018. [Invited talk]

OTHER ACTIVITIES

- Ongoing Reviewer for top journals including Geoscientific Model Development, JGR: Atmospheres, Geophysical Research Letters, Environmental Research Letters, Atmospheric Chemistry and Physics, and Urban Environments
- Jan. 2026 - present Early Career Network Committee member, Faculty of Science, Monash University
- Dec. 2025 Research visiting, Max Planck Institute for Meteorology, Dr Hans Segura
- Nov. 2025 Research visiting, the Arctic University of Norway (UiT), Dr Nikolas Aksamit
- Jun. 2025 - present Early Career Researcher Committee member, School of Earth Atmosphere and Environment, Monash University
- Sep. 2025 Guest lecturer for fire modelling, EAE5053 Advanced Spatial Analysis and Modelling
- Feb. - Apr. 2025 Supervision for Summer Project
- Jul. 2024 - Mar. 2025 Early Career Researcher Committee member, CLEX
- Aug. 2022 - Dec. 2023 Research consultancies:
- Wind observations for Kea Aerospace
 - Extreme wind event investigation for the 2022 Lake Ōhau fire
 - Fire weather index and future fire risk projection in New Zealand
- Oct. 2023 Guest lecturer, GEOG310 Weather Systems
- Aug. - Oct. 2023 Supervision for undergraduate research projects
- Assessing the impacts of the 2019-2020 Australian bushfires
 - 3D visualisation of Doppler wind lidar observations
- Jun. 2023 Invited speaker, Clean Air Society of Australia and New Zealand (CASANZ) WRF workshop
- Nov. - Dec. 2021 Research Assistant (consultancy), ash hazard and impact for the Scott Base redevelopment, Antarctica New Zealand
- Jul. - Oct. 2021 Research Assistant (consultancy), air pollution modelling for Whangarei, New Zealand
- Jun. 2021 Organiser, research group writing retreat at the Cass field station
- Nov. 2020 Organiser, leading a two-day WRF workshop for the MetSoc student session
- Aug. 2020 and 2022 Guest Lecturer and Teaching Assistant, PHYS330 Climate Modelling
- Apr. - Jun. 2020 Research Assistant (consultancy), WRF simulations of Early Valley Road wind conditions during the Port Hills 2017 fires for DLA Piper New Zealand
- Aug. - Sep. 2019 Research visiting, Karlsruhe Institute of Technology (KIT), Campus Alpin, Institute of Meteorology and Climate Research, Department of Atmospheric Environmental Research (IMK-IFU), Dr Basit Khan
- Jul. 2019 Lab demonstrator, University of Canterbury Open Day
- May 2019 and 2023 Outreach activity, Hororata Night Glow Festival

KEY SKILLS

- Atmospheric Dynamics, Extreme Rainfall, and Boundary Layer Meteorology
- Field work and field data management, including radiosonde, lidar, and sonic-anemometers
- Python Programming: Adept at developing and maintaining scientific software for numerical models
- Atmospheric Numerical Modelling: Experienced in WRF and PALM (Large Eddy Simulation)
- High-Performance Computing Clusters: Compile models and conduct simulations (> 2,000 CPUs)
- Geospatial Data Analysis and Visualization: QGIS, ArcGIS and Python
- Big Data Processing and Analysis: Adept in handling and analysing large datasets (> 50 TB)

LANGUAGE PROFICIENCY

- English: superior
- Chinese (Mandarin): native